

(b) The drug L-DOPA is used in tablet form to treat Parkinson's disease.

This drug is converted to dopamine by an enzyme found in the digestive system and the brain.

Carbidopa is an inhibitor of the enzyme found in the digestive system.

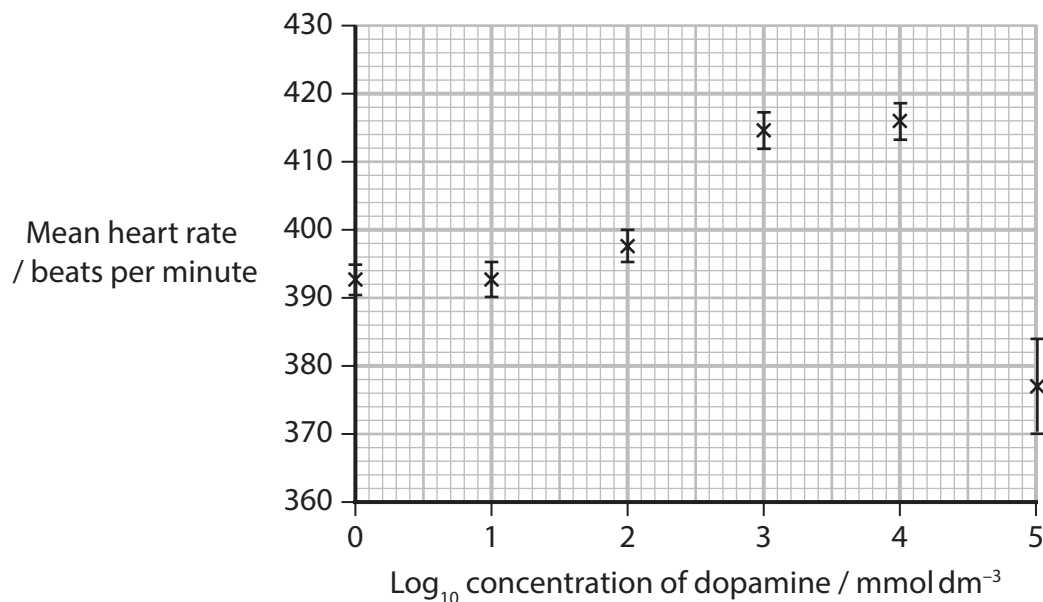
Suggest why carbidopa is given to patients when they are treated using L-DOPA in tablet form.

(2)

- (c) In an investigation, different concentrations of dopamine were given to *Drosophila* (fruit flies).

The mean heart rates of groups of 10 *Drosophila* were recorded at different concentrations of dopamine.

The results are shown in the graph.



Comment on the results of this investigation.

(4)

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(Total for Question 4 = 10 marks)